

Trading Scope for Credibility in Difference-in-Differences

Parush Arora*

Abhishek Chand†

Abstract

When parallel trends fails for some treated cohorts but not others, the average treatment effect on the treated (ATT), an average over all of them, is exactly the target that becomes hard to recover. We propose changing the estimand rather than defending it. The credible-subpopulation local ATT (LATT) is the effect for the subpopulation of cohorts whose parallel trends is credible, and it is point-identified under parallel trends for the selected cohorts alone, a weaker requirement that can hold when the ATT's fails. It is estimated by reweighting standard group-time effects toward those cohorts, and paired with honest sensitivity bounds on the residual violation that a pre-trend screen cannot rule out. The method's advantage grows with how informative pre-trends are about post-treatment violations, as simulations confirm. In an application, a significantly positive pooled estimate of the shale boom's effect on local house prices proves to rest on cohorts already trending before onset, and the credible subpopulation reveals no effect.

Keywords: Difference-in-differences, Staggered adoption, Parallel trends, Robust inference, Sensitivity analysis, Partial identification.

JEL classification: C1, C23, C51.

*Department of Economics, Ashoka University.

†Data Scientist, PwC.

1 Introduction

Difference-in-differences is among the most widely used research designs in empirical economics, and its credibility rests on the parallel trends assumption, that absent treatment treated and comparison units would have evolved in parallel. Researchers assess it by testing for pre-treatment differences in trends, but that practice is fragile. Pre-trends tests are often underpowered against economically meaningful violations (Bilinski and Hatfield, 2020; Freyaldenhoven et al., 2019; Kahn-Lang and Lang, 2020; Roth, 2022), conditioning on passing them induces a pre-test selection bias (Roth, 2022), and even a clean pre-period does not guarantee a clean post-period, since nothing prevents a confound from arriving at the moment of treatment (Kahn-Lang and Lang, 2020).

A growing literature, surveyed in Roth et al. (2023), has responded by relaxing parallel trends itself. Building on Manski and Pepper (2018), Rambachan and Roth (2023) abandon exact parallel trends and instead bound how different the post-treatment violations can be from the observed pre-trends, reporting a uniformly valid confidence set for the ATT in place of a point estimate. A Bayesian variant (Kwon and Roth, 2024) places an empirical-Bayes prior on the violation δ to sharpen the set into a posterior mean and credible interval, and de Chaisemartin (2026) instead ranks the post-treatment difference-in-differences against the pre-treatment ones. All keep the ATT as the target and use the pre-trends to extrapolate the violation across periods.

The ATT is where a structural limitation appears. When parallel trends fails for some cohorts but holds for others, the ATT, an average over all treated cohorts including the offending ones, is precisely the object that is hard to recover. A point estimate of it is then biased, and the sensitivity set, though honest, can be wide and uninformative, because it must accommodate the worst offenders in the average.

This paper takes a different route. When the population target is not credibly identified, we retreat to a subpopulation where it is. This is the logic of the local average treatment effect (Imbens and Angrist, 1994), of the optimal-subpopulation average treatment effect under limited overlap (Crump et al., 2009), and of overlap weighting (Li et al., 2018). In each case one trades the original estimand for a credibly-identified effect on a selected subpopulation, and defends the trade by characterizing whom the subpopulation comprises.

We develop the staggered-DiD instance of this principle. The credible-subpopulation LATT is the weighted average of group-time effects restricted to cohorts whose parallel trends is credible, and it is point-identified under parallel trends for the selected cohorts alone, a weaker requirement than the ATT's that can hold when the latter fails, since the excluded cohorts' violations do not enter it. Estimation is a transparent reweighting of standard group-time effects (Callaway and Sant'Anna, 2021), so the method applies to any heterogeneity-robust estimator that delivers them.¹ Credibility is assessed from the pre-trends, making the reweighting a data-dependent selection, whose inference

¹These estimators (Callaway and Sant'Anna, 2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D'Haultfoeulle, 2020) repair the aggregation failure of two-way fixed effects under staggered adoption (Goodman-Bacon, 2021), whose implicit weights can turn negative under treatment-effect heterogeneity. That repair concerns aggregation, not identification, since they maintain parallel trends and are biased when it fails, which is the failure we address.

we treat with the sensitivity machinery of Rambachan and Roth (2023).

The method thus occupies a distinct point on a frontier of credibility, precision, and scope. Against a point estimate of the ATT it buys credibility by narrowing the target, and against a HonestDiD set for the ATT it buys a sharper, point-identified answer. Its advantage is governed by how informative pre-trends are about post-treatment violations. When they are uninformative, selection buys nothing and the assumption must rest on economic context, and there the relative-magnitudes restriction is self-undermining, reporting false precision when pre-trends are flat where a fixed, researcher-set bound is not.

The remainder of the paper is organized as follows. Section 2 fixes the staggered-DiD setup, defines the credible-subpopulation LATT, and develops its identification, estimation, and sensitivity-based inference. Section 3 reports the simulation study, Section 4 applies the method to the local house-price effect of the shale boom, and Section 5 concludes.

2 The credible-subpopulation LATT

2.1 Setup

We adopt the staggered adoption framework of Callaway and Sant’Anna (2021). There are periods $t = 1, \dots, T$ and units i indexed by their first treatment date $G_i \in \{2, \dots, T\} \cup \{\infty\}$, where $G_i = \infty$ denotes never-treated. Let $Y_{it}(g)$ denote the potential outcome in period t if unit i is first treated in period g , and $Y_{it}(\infty)$ the never-treated potential outcome. The building-block causal parameter is the group-time average treatment effect on the treated,

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g]. \quad (1)$$

Summary parameters are weighted aggregations $\theta = \sum_g \sum_t w(g, t) \text{ATT}(g, t)$ with researcher-chosen weights (Callaway and Sant’Anna, 2021), of which the overall ATT and the event-study path are special cases.

It is convenient to work with the event-study representation and the decomposition of Rambachan and Roth (2023). Let $\hat{\beta} = (\hat{\beta}'_{\text{pre}}, \hat{\beta}'_{\text{post}})'$ be an asymptotically normal vector of event-study coefficients estimated by any of the heterogeneity-robust procedures above, with $\sqrt{n}(\hat{\beta} - \beta) \rightarrow \mathcal{N}(0, \Sigma)$. The estimand decomposes as

$$\beta = \tau + \delta, \quad \tau_{\text{pre}} = 0, \quad (2)$$

where τ collects the causal effects (zero before treatment, by no anticipation) and δ is the differential trend between treated and comparison groups that would have occurred absent treatment. Parallel trends is the restriction $\delta_{\text{post}} = 0$, under which $\beta_{\text{post}} = \tau_{\text{post}}$, and a pre-trends test is a test of $\delta_{\text{pre}} = 0$.

We will require the analogous objects at the cohort level. Write δ_g for the differential trend of cohort g and say that parallel trends holds for g if $\delta_{g, \text{post}} = 0$. A key feature of applications is that this may hold for some cohorts and fail for others. Defending the timing of treatment as unrelated

to the outcome path is already delicate in a two-group design, and staggered adoption raises the burden rather than lowering it, since each cohort's timing must be defended on its own, one cohort's adoption perhaps as good as randomly timed while another's coincides with a confounding shock.

Finally, we distinguish two regimes for how cohorts are judged credible, because they carry different inferential guarantees. Under ex-ante (or independent) selection, credibility is decided from pre-determined covariates, institutional knowledge, or an independent data split, so that the selected set is a fixed S statistically independent of the estimation errors in $\hat{\beta}$. Under data-driven selection, credibility is assessed from the realized pre-trends $\hat{\beta}_{\text{pre}}$, so that the selected set $\hat{S}$ is random and depends on the estimation sample, with a deterministic population counterpart S^* , the set a population pre-trend statistic would select. The identification argument below holds for any set held fixed, and so identifies the random target $\theta_{\hat{S}}$ conditional on the realized $\hat{S}$. Identifying the fixed population parameter θ_{S^*} instead requires $\hat{S}$ to recover S^* , a selection-consistency condition we return to under inference (Proposition 4). The distinction is thus minor for identification but central for inference, and we treat it explicitly below.

2.2 Estimand and identification

Let S denote a set of cohorts judged to have credible parallel trends, produced by a selection rule R that we specify below. We define the target as the treatment effect for that subpopulation.

Definition 1 (Credible-subpopulation LATT). *For cohort weights $w_g > 0$ (e.g. proportional to cohort size) and a within-cohort aggregation $a(\cdot)$, a linear functional of a cohort's post-treatment vector (a fixed event-time, selecting one coordinate, or an average over post-treatment periods), the credible-subpopulation local ATT, defined for any nonempty S , is*

$$\theta_S = \frac{\sum_{g \in S} w_g \text{ATT}(g, \cdot)}{\sum_{g \in S} w_g}, \quad \text{ATT}(g, \cdot) = a(\tau_{g, \text{post}}), \quad (3)$$

the aggregated causal effect of cohort g .

The value of the target is that it is identified under a weaker condition than the ATT.

Proposition 1 (Identification). *If parallel trends holds for every $g \in S$, i.e. $\delta_{g, \text{post}} = 0$ for all $g \in S$, then θ_S is point-identified by the reweighted aggregated group-time effects, $\theta_S = (\sum_{g \in S} w_g a(\beta_{g, \text{post}})) / \sum_{g \in S} w_g$, and this holds irrespective of whether $\delta_{g, \text{post}} \neq 0$ for cohorts $g \notin S$.*

Proof. For $g \in S$, $\beta_{g, \text{post}} = \tau_{g, \text{post}} + \delta_{g, \text{post}} = \tau_{g, \text{post}}$ by the hypothesis and (2), so $a(\beta_{g, \text{post}}) = a(\tau_{g, \text{post}}) = \text{ATT}(g, \cdot)$ by linearity of a . The reweighted sum over S therefore equals the weighted average of the aggregated causal effects, which is θ_S by definition. Cohorts outside S receive zero weight, so their differential trends do not enter. $\square$

Proposition 1 is elementary, but it is the paper's central point. Changing the target to the credible subpopulation weakens the identifying requirement from parallel trends on all cohorts to

parallel trends on S , which can hold when the former fails, at the cost of a narrower question whose subpopulation S must then be characterized. The weakening is genuine but conditional, and two things must be kept separate. The identifying content is the substantive restriction $\delta_{g,\text{post}} = 0$ for $g \in S$, an assumption about post-treatment trends. The flatness screen of Section 2.3 is a statistical classification rule that supplies evidence for that restriction from the pre-period, not the restriction itself, since a cohort flat before treatment may still drift after it. Point identification of the causal θ_S thus holds only when the restriction does, and the screen cannot certify it. We therefore treat the point-identified θ_S as a benchmark and pair it with the honest sensitivity bounds developed below, which bound the residual post-treatment violation the screen leaves behind and, in doing so, return a set rather than a point when that violation is nonzero.

The construction uses staggered timing only to supply the groups. What it requires is a partition of the treated population into groups whose parallel trends can each be assessed. Adoption cohorts are the natural groups, and their credibility heterogeneity comes for free from the variation in when and why units adopt. In a non-staggered design the same construction applies with groups defined by covariates or geography, provided the groups differ in credibility and there are enough pre-periods to estimate each group’s pre-trend rather than chase noise. The groups should accordingly be subpopulations large enough to assess, not individual units, whose pre-trends are too noisy to select on without reviving the pre-test bias in its most severe form.

2.3 Selection and estimation

The selection rule R maps the data to the set S , and its form fixes both which subpopulation θ_S describes and the inferential guarantees available. It takes two forms, matching the two regimes of Section 2.1. Under ex-ante selection, S is fixed before estimation from pre-determined covariates, institutional knowledge of which cohorts were plausibly confounded, or an independent data split. Under data-driven selection, credibility is read from the estimated pre-trends, and a cohort enters S when its estimated pre-trend is close to flat,

$$S = \{ g : \max_{e < 0} |\hat{\beta}_{g,\text{pre}}(e)| \leq c \}, \quad (4)$$

for a threshold c .

This is the direct reading of parallel trends. The assumption is that the treated-comparison difference would have stayed constant absent treatment, so a cohort is credible when its pre-treatment difference is close to constant, and the screen keeps exactly those. It is posed in the same currency as the honest inference of the next subsection, which bounds how far the post-treatment difference could have wandered from flat, so selection and inference test one object, the level of the parallel-trends violation.²

²A researcher willing to trust that a straight pre-trend would have continued linearly can relax flatness to approximate linearity, screening instead on the pre-trend’s curvature (its second difference) and pairing it with the smoothness sensitivity class of Rambachan and Roth (2023) and a linear-extrapolation estimator (Armstrong and Kolesár, 2018). This buys the steep-but-straight cohorts that flatness discards, at the cost of the extrapolation assumption. The principle is the same in either currency. The screen should be posed in whichever functional the

The threshold trades scope against credibility, with a lax c retaining more cohorts but admitting larger residual violations and a strict c the reverse. Because rule (4) reads S off the noisy estimates $\hat{\beta}_{\text{pre}}$, the selected set is random and dependent on the estimation sample, which is the source of the pre-test bias below and of the inference subtleties that follow it.

We do not recommend committing to a single c . The sensitivity interval of Section 2 is applied to whichever set c selects, and its bound M must dominate that set's residual *post*-treatment violation, so the effect of c on the required M runs through how far screening on flat pre-trends also flattens post-trends. When pre-trends are informative about post-treatment violations, a stricter c yields a cleaner set that a smaller M suffices to cover, while a laxer c buys scope at the cost of a larger M , so the threshold trades scope for precision rather than credibility for nothing. When they are not, tightening c need not shrink the post-treatment violation and the required M can be nonmonotone in c . Either way the honest object to report is the estimand together with its interval traced as a function of c , the selection analogue of the breakdown curve we report for M . A reader then sees the scope-credibility-precision frontier directly and need not trust a single cutoff or a presumed monotone mapping. When a single value is nonetheless wanted, we recommend calibrating it to negligibility rather than to significance, setting c to the largest pre-trend that would bias θ_S by less than a stated tolerance, in the spirit of the non-inferiority tests of Bilinski and Hatfield (2020), rather than to a critical value of a pre-trends test, which would reinherit the low power that motivates the paper. The application of Section 4 reports such a path.

The estimator $\hat{\theta}_S$ simply reweights, over the selected cohorts, the sample group-time effects of whichever heterogeneity-robust estimator one uses, and nothing below depends on that choice.

Proposition 2 (Consistency and efficiency). *Under ex-ante selection and the regularity conditions of the underlying group-time estimator, $\hat{\theta}_S$ is consistent for θ_S . If in addition parallel trends holds for all cohorts, $\hat{\theta}_S$ is consistent for the ATT when either treatment effects are homogeneous or S comprises all cohorts, and otherwise for the selected-cohort LATT, which differs from the ATT by the gap between the selected- and full-population weighted averages of the cohort effects. In either case there is an efficiency loss relative to estimators that use all cohorts.*

Under data-driven selection the rule (4) couples S to the estimation errors: a confounded cohort whose pre-trend noise happens to fall below c is admitted, and its differential trend enters $\hat{\theta}_S$. This is a pre-test selection bias in the sense of Roth (2022), with a definite source and a definite fate.

Proposition 3 (Pre-test bias). *Under data-driven selection, $\hat{\theta}_S$ carries a finite-sample bias equal to the weighted average of the admitted confounded cohorts' post-treatment differential trends, times their probability of admission. Whether it vanishes turns on detectability. Call a confounded cohort detectable if its population pre-trend statistic exceeds the threshold, $\max_{e < 0} |\beta_{g,\text{pre}}(e)| > c$, and undetectable if it is flat in the pre-period, $\max_{e < 0} |\beta_{g,\text{pre}}(e)| \leq c$, yet violates parallel trends in the post-period. As per-cohort information grows, a detectable cohort's admission probability tends to*

sensitivity class bounds, flatness with the level bound and curvature with the smoothness class, since level and curvature rank cohorts differently and select different sets.

zero and its contribution to the bias vanishes, whereas an undetectable cohort's admission probability tends to one and its contribution persists at every sample size. Hence $\hat{\theta}_S$ is consistent for θ_S if and only if every confounded cohort is detectable, and otherwise retains an asymptotic bias equal to the residual violation of the undetectable cohorts, which is the identification gap that the honest inference of the next subsection bounds.

2.4 Honest inference on the selected set

Selecting credible cohorts does not by itself deliver parallel trends, since a cohort with a flat pre-trend may still have a nonzero post-treatment differential trend, and selection removes gross offenders but not subtle residual violations. We therefore accompany the point estimate with the sensitivity analysis of Rambachan and Roth (2023), applied to the selected-cohort aggregate, imposing a restriction $\delta_{S,\text{post}} \in \Delta$ on the residual differential trend and reporting the implied confidence set for θ_S . Matching the flatness criterion of the screen, we adopt the level bound $\Delta^{\text{Level}}(M) = \{\delta : |\delta_{\text{post}}(e)| \leq M \forall e\}$, which asks how far the treated-comparison difference could have wandered from its flat pre-treatment level, absent treatment. The estimator is then the raw reweighted post-treatment average, with no extrapolation, and the associated fixed-length confidence interval (FLCI) is near-optimal (Armstrong and Kolesár, 2018). Because M is measured in the outcome's own units, the breakdown value M^* reads directly as the size of post-treatment parallel-trends violation that would just overturn the stated conclusion.

Validity depends on the selection regime. Under ex-ante selection S is independent of $\hat{\beta}$, so the uniform validity of Rambachan and Roth (2023) applies verbatim. Under data-driven selection S is random, and two issues arise. The first is post-selection inference. When cohorts are well separated from the threshold, selection is asymptotically harmless.

Proposition 4 (Selection consistency and oracle coverage). *Suppose the statistic on which selection is based is consistent for its population counterpart, that the following separation condition holds, that there exists $\epsilon > 0$ such that, for every cohort, the population value of that statistic lies at distance at least ϵ from the threshold c , and that the population credible set's residual violation satisfies the maintained restriction $\delta_{S^*,\text{post}} \in \Delta^{\text{Level}}(M)$. Then $P(\hat{S} = S^*) \rightarrow 1$, where S^* denotes the population credible set, and the FLCI computed on the estimated set $\hat{S}$ has the same asymptotic coverage as the oracle interval computed on S^* . Selection consistency delivers this equivalence to the oracle interval, and the maintained restriction delivers the oracle interval's nominal coverage of the causal target, so the two together give nominal coverage.*

Proof. By consistency of the selection statistic together with the separation condition, the probability that any given cohort is misclassified tends to zero. Since the number of cohorts is finite, a union bound gives $P(\hat{S} \neq S^*) \rightarrow 0$. On the complementary event the estimator, its covariance estimate and the resulting interval coincide exactly with their oracle counterparts computed on the non-random set S^* . The two sequences therefore share a limiting distribution, and the coverage of the interval computed on $\hat{S}$ converges to that of the oracle interval, which is nominal by the validity of

Rambachan and Roth (2023) applied to a fixed set whose residual violation lies in $\Delta^{\text{Level}}(M)$. $\square$

Two caveats attach to Proposition 4. First, the separation condition is untestable, and is nearly the parallel trends question in disguise, since to assert that every cohort is plainly credible or plainly not is close to asserting one already knows which cohorts are credible, so the result relocates the identifying content rather than supplying it. Second, the guarantee is pointwise, not uniform. Leeb and Pötscher (2005) show that a post-selection estimator’s distribution cannot be estimated uniformly consistently, so naive post-selection intervals lack uniform validity (Leeb and Pötscher, 2008). The obstruction is confined to local-to-threshold configurations, where the selection statistic drifts toward the boundary at the sampling rate. Away from the boundary, selection is asymptotically deterministic and inference behaves conventionally. A researcher unwilling to assume separation can sidestep both caveats by selecting on information independent of the estimation errors, such as pre-determined covariates, a sample split, or randomized (data-carving) selection (Fithian et al., 2014). These buy validity through independence between selection and estimation rather than correct selection, at some cost in efficiency.

The second issue is that the bound M must accommodate a random selected set. The FLCI at M is valid only if the selected set’s residual violation lies in $\Delta^{\text{Level}}(M)$. Since that violation is itself random, unconditional coverage requires M to dominate it with high probability.

Lemma 1 (Calibration to the random selected set). *Under data-driven selection the selected aggregate’s residual violation is random, and the FLCI’s unconditional coverage is governed by the joint distribution of that violation, the selection event, and the sampling error. When the violation’s dispersion is non-negligible relative to the FLCI’s sampling slack, setting M to the mean residual violation undercovers, because the realized violation exceeds M in a nontrivial fraction of samples, and restoring nominal unconditional coverage requires M to exceed the mean and approach an upper quantile of the residual-violation distribution, with the sampling slack providing a partial buffer. When the slack instead dominates the dispersion the buffer alone can suffice.*

The practical implication, in the regime where the violation’s spread is the binding consideration, is that M should be calibrated to the residual violation of a typical selected set, not to its mean.

Finally, we caution against one otherwise-natural choice of Δ . The relative-magnitudes restriction $\Delta^{RM}(\bar{M})$ of Rambachan and Roth (2023), following Manski and Pepper (2018), bounds post-treatment violations by $\bar{M}$ times the observed maximal pre-treatment violation, so its identified-set half-width is proportional to the observed pre-trend magnitude. When pre-trends are uninformative about post-treatment violations, the observed pre-trends are small while the true violation need not be, so the restriction produces intervals whose width tends to zero even as the bias does not, yielding undercoverage precisely in the regime where our method’s value is in question. A fixed restriction such as $\Delta^{\text{Level}}(M)$, whose bound is set by the researcher rather than read off the observed pre-trends, does not share this defect.

3 Simulation study

3.1 Design

We study the method in two environments. In the reduced-form model we draw event-study coefficients directly from their asymptotic distribution, $\hat{\beta} \sim \mathcal{N}(\tau + \delta, \Sigma)$ with Σ known, which isolates the identification and inference mechanics in the setting where the underlying theory is exact. In the panel model we generate individual outcomes for a staggered design and estimate both the coefficients and their covariance from the data, which confirms that the conclusions survive realistic estimation. Unless stated otherwise, results are averages over eight to twenty thousand replications.

The cohort structure follows a common template. Some cohorts are clean, with $\delta_g = 0$, and the remainder confounded, with a nonzero differential trend, and Table 1 gives the count and clean-confounded split used in each exercise. Every cohort is given the same true treatment effect, normalized to one. This normalization is deliberate, making the true ATT and the true LATT both equal to one, so that any departure of an estimator from one is attributable to a violation of parallel trends rather than to treatment-effect heterogeneity. It therefore isolates the identification problem that the paper is about.

Throughout, each cohort carries a four-period pre-treatment and four-period post-treatment event study with the reference period normalized to zero. A confounded cohort carries a level violation, its treated-comparison difference shifting by $V > 0$ in the post-period and by ϕV in the pre-period, for a scalar $\phi \geq 0$ we call the informativeness parameter and that is the master axis of the study. When $\phi = 0$ the confound is flat before treatment and no pre-trend rule can detect it, and as ϕ grows the pre-period shift announces it and it becomes detectable. Because V is the post-treatment violation it is directly comparable to the level bound M of $\Delta^{\text{Level}}(M)$. In the controlled coverage exercise below the aggregate being tested carries this violation exactly, which makes the boundary statement “covers if and only if $M \geq V$ ” checkable. More generally, coverage requires only that M dominate the selected aggregate’s violation, which for a set mixing clean and confounded cohorts is weakly below the largest single-cohort V , so $M \geq V$ is sufficient and is necessary only in the worst case of an all-confounded set. This also makes ϕ the operational version of the informativeness of pre-trends discussed in Section 2.

The selection rule is the flatness screen, posed in the same currency as the honest inference, so cohort g is retained if its estimated pre-trend is close to flat, $\max_e |\hat{\beta}_{g,\text{pre}}(e)| \leq c$. The estimator of the ATT averages the post-treatment effect over all cohorts, as a heterogeneity-robust estimator would, and the LATT averages it over the selected cohorts only. Because the screen reads the selected set off the same noisy pre-trends, the selection is genuinely data-dependent in the sense of Section 2, and the pre-test bias of Proposition 3 is operative.

Table 1: Simulation parameters. Every exercise uses a four-period pre and four-period post event study with reference period zero, equal cohort weights, and true effect one for every cohort. The reduced-form model draws coefficients directly, $\hat{\beta} \sim \mathcal{N}(\tau + \delta, \Sigma)$ with within-cohort AR(1) covariance $\Sigma_{ij} = s^2 \rho^{|i-j|}$. The panel model draws individual untreated outcomes $Y_{it}(0) = \delta_g(e) + \epsilon_{it}$, $\epsilon \sim \mathcal{N}(0, \sigma_\epsilon^2)$, for 500 treated units per cohort and 1500 shared controls over $T = 14$ periods, and estimates the coefficients and their covariance from the data.

	Estimand (Table 2)	Scope (Fig. 1)	Honest cov. (Table 3, Fig. 2)	Panel (Sec. 3.5)
Cohorts (clean/confounded)	6 (4/2)	12 (6/6)	—	6 (3/3)
Post-treatment violation V	0.8	0.6	0.3, 0.6	0.6
Informativeness ϕ	1.0	[0, 1.5]	—	[0, 1.5]
Screen threshold c	0.40	0.40	—	0.40
Noise	$s=0.03\text{--}0.40$	$\sigma=0.20$	$\sigma_{\text{agg}}=0.10$	$\sigma_\epsilon=0.5$
Pre/post correlation ρ	0.5	0.5	—	estimated
Replications	20,000	8,000	4,000	2000–3000

3.2 Recovering a clean effect where the ATT is biased

We first illustrate Proposition 1 in the simplest configuration, with a subset of cohorts violating parallel trends in the post-period. Table 2 reports the result. The estimator of the ATT is biased by +0.267, since it averages over all cohorts, so the offending cohorts’ differential trends enter it directly. The LATT lands at 1.000, on the truth, because the selection removes those cohorts from the average and, by Proposition 1, their violations then do not enter the estimand at all.

The lower panel of the table sweeps the sampling noise s , and is the more informative comparison. As precision improves the LATT’s bias falls monotonically from +0.013 to zero, and this residual is the pre-test bias of Proposition 3, which vanishes as the selection rule becomes reliable. Two sources could in principle generate it, a noisy screen that admits genuinely confounded cohorts, which would persist even under independent selection, and the correlation between a cohort’s pre- and post-treatment sampling errors, which distorts the retained cohorts’ estimates only under same-sample selection. The final column isolates the two by selecting on an independent pre-trend draw. It tracks the full-sample column almost exactly, so the residual is the first source, admission of confounded cohorts, and the second is negligible here because the two-sided flatness screen conditions the retained estimates symmetrically. The ATT estimator’s bias, by contrast, is frozen at +0.267 regardless of precision. The contrast is the paper’s basic point in miniature, the LATT facing a noise problem, which more data cures, whereas the ATT estimator faces a wrong-target problem, which no amount of data cures.

3.3 The scope condition

Figure 1 traces the method across the informativeness axis ϕ described in Section 3.1, and its four panels should be read together.

The upper-left panel plots the bias of the two point estimators against the truth. The bias of the

Table 2: The estimand demonstration (reduced-form model). True effect = 1 for all cohorts. In the lower panel LATT (full) selects on the same pre-trends used for estimation, while LATT (split) selects on an independent draw. Their near-equality shows the residual bias is admission of confounded cohorts, not same-sample conditioning.

	ATT	LATT (full)	LATT (split)
Point estimate	1.267	1.000	
Bias vs. truth	+0.267	+0.000	
Residual LATT bias as sampling noise s falls			
$s = 0.40$	+0.265	+0.013	+0.014
$s = 0.20$	+0.267	-0.001	-0.000
$s = 0.12$	+0.267	-0.000	-0.000
$s = 0.03$	+0.267	+0.000	+0.000

ATT estimator is a flat line at +0.30, since it never consults pre-trends, so its performance cannot depend on how informative they are. The LATT’s bias begins on top of that line at $\phi = 0$ and falls to zero as pre-trends become informative. The vertical gap between the two curves is therefore exactly the method’s advantage, and reading it off from left to right gives the scope condition, with the advantage nil when pre-trends carry no information about post-treatment violations and maximal when they carry a great deal.

That the two curves coincide at the left edge is not a defect but an honest reflection of the fact, argued in Section 2, that selecting on uninformative pre-trends buys nothing.

The upper-right panel explains why the LATT’s bias falls, by plotting the identification gap of the selected set, the average post-treatment violation among the cohorts that survive selection. It tracks the LATT’s bias almost exactly, falling from 0.30 to essentially zero. This confirms that the LATT’s remaining bias is not an artifact of the estimator but simply the residual violation that selection has failed to remove. The lower-right panel gives the mechanism behind both, the average number of confounded cohorts that slip through the screen falling from 5.09 of six when the confound is invisible to essentially zero when it is plainly visible.

The lower-left panel turns to inference and reports the coverage of a point-based confidence interval for the causal target, computed both in the naive way and by sample-splitting. Both collapse at low informativeness and recover only at high informativeness. The comparison is diagnostic, since sample-splitting removes any contamination from data-dependent selection by construction, so the fact that it collapses just as badly establishes that the failure is not selection distortion but the identification gap of the upper-right panel, since the surviving cohorts genuinely still violate parallel trends. This is what motivates the sensitivity bounds of the next subsection, since no purely inferential fix can repair a violation of the identifying assumption.

(At the right edge the naive interval slightly outperforms the split one. This is simply because splitting discards half the data, and by that point selection is nearly deterministic, so there is no contamination left to correct.)

Master-axis sweep: everything is governed by pre-trend informativeness
($\theta=1$ for all cohorts, so true ATT = true LATT = 1; flatness screen)

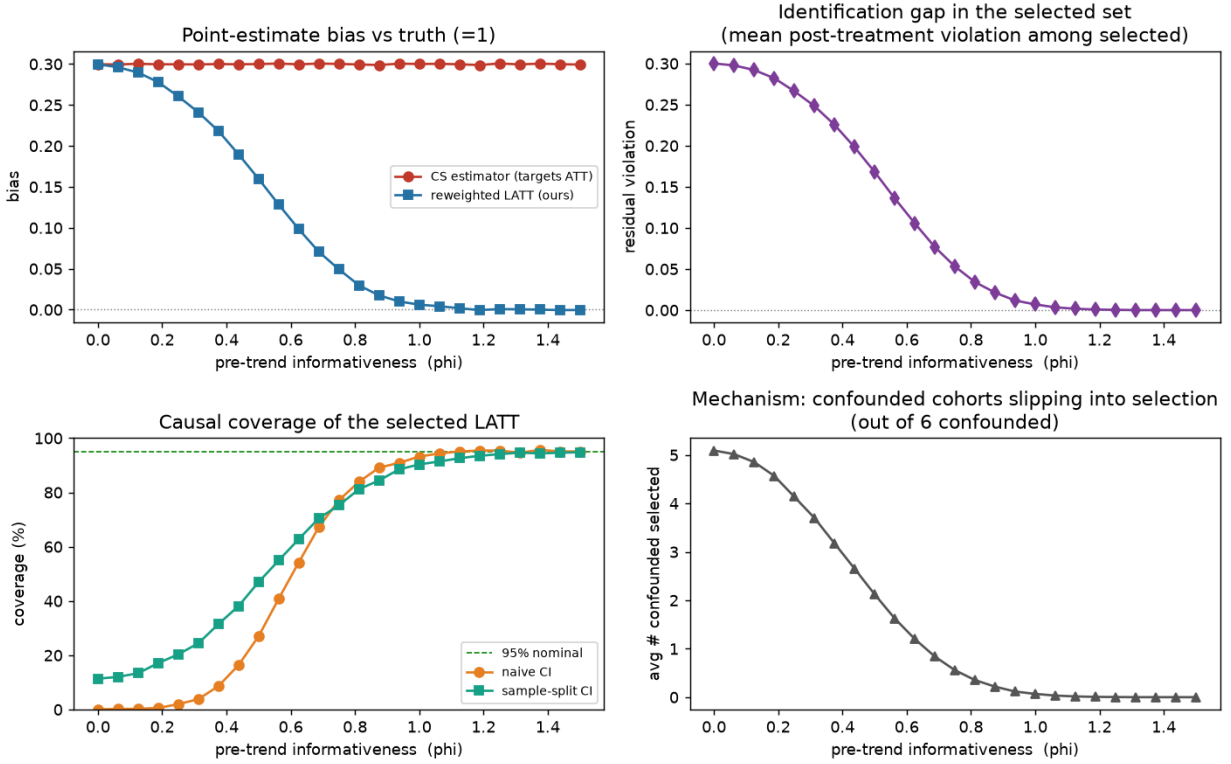


Figure 1: Scope condition. Everything is governed by pre-trend informativeness. The ATT estimator’s bias is constant, and the LATT’s advantage is the gap between the lines. Causal-coverage collapse at low informativeness is an identification gap (surviving cohorts still violate parallel trends), not selection distortion, since sample-splitting collapses just as badly.

3.4 Honest inference restores coverage

We now validate the level-bound FLCI of Section 2 applied to the selected-cohort aggregate, using the level violation of Section 3.1 whose size V is directly comparable to the assumed bound M .

Table 3 is the validity check. Read down the rows. When the assumed bound equals the true violation ($V = M$) the FLCI covers at 95.0%, that is, at its nominal level. When the bound is generous ($M > V$) it covers conservatively at 100%. When the bound is violated ($M < V$) coverage collapses. This is precisely the behavior an honest sensitivity interval should display. It is valid exactly on the range of violations it claims to accommodate, and it fails visibly outside it, rather than failing silently.

The final column records that the naive point interval collapses to near-zero coverage the instant any violation is present, because it is centered on a biased estimate with a width reflecting sampling noise alone.

Figure 2 repeats the exercise across a continuum of violation sizes and adds the practical output.

Table 3: FLCI coverage of the causal target (reduced-form model), for an aggregate whose violation equals V , valid iff $M \geq V$.

True violation V	Assumed M	FLCI coverage	Point-estimate coverage
0.30	0.30 ($= V$)	95.0%	15.9%
0.30	0.60 ($> V$)	100%	14.9%
0.60	0.30 ($< V$)	9.3%	0.0%
0.60	0.60 ($= V$)	95.0%	0.0%

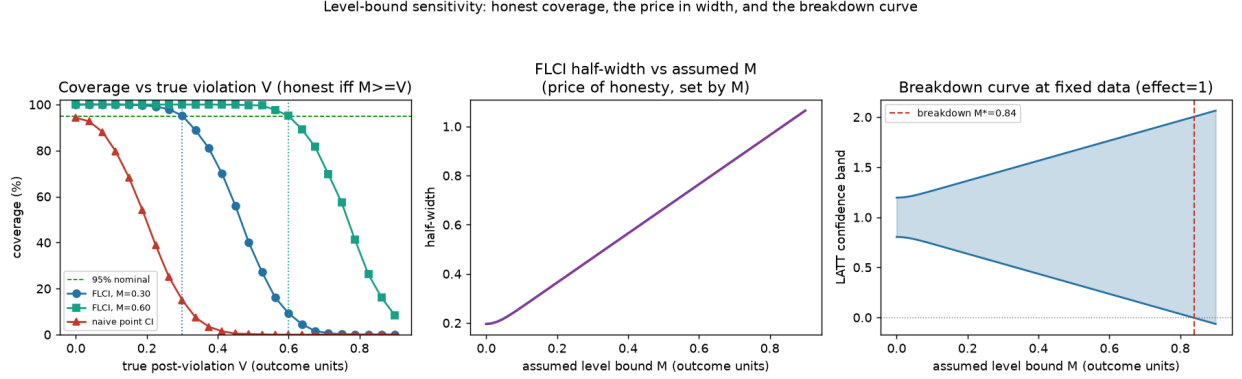


Figure 2: Sensitivity bounds restore honest coverage. The point estimate collapses as the violation grows. The FLCI is honest exactly while $M \geq V$. Interval width is the price of honesty, set by M . The right panel is the sensitivity/breakdown curve, with M^* in outcome units.

In the left panel the point interval falls away rapidly as the violation grows, while each FLCI holds its nominal level until the true violation reaches its own assumed bound, marked by the vertical dotted lines, and declines thereafter. The middle panel records the price. Interval half-width is governed by the assumed M rather than by the realized violation, so honesty about a wider class of violations is paid for in width, immediately and proportionately.

The right panel is what a practitioner would actually report. Holding the data fixed, it traces the confidence band as the assumed bound is relaxed and identifies the breakdown value M^* at which the band first admits the truth. Because M is in the outcome's own units, this quantity answers the question a reader of a DiD paper wants answered directly, how large a post-treatment parallel-trends violation one must be willing to countenance before the stated conclusion no longer follows. And because the bound is fixed by the researcher rather than read off the pre-trends, the exercise remains informative even when pre-trends are uninformative, in direct contrast to the relative-magnitudes restriction, whose bounds would shrink toward zero in exactly that case and deliver false precision.

3.5 Selection, calibration, and panel confirmation

Two questions remain for the data-driven regime, whether choosing the selected set from the same data distorts the sensitivity interval, and how to choose M when the selected set, and hence its

residual violation, is random. On the first, coverage against M for the full-data and sample-splitting procedures crosses the nominal level at nearly the same bound ($M \approx 0.048$ and 0.056), so data-dependent selection introduces no gross distortion and splitting buys little, consistent with the pointwise reading of Leeb and Pötscher (2005). On the second, the residual violation of the selected aggregate is not a constant but a spread, multi-modal random variable across replications, running from zero to about 0.06 with a thin tail to 0.11, each mode a different number of borderline cohorts surviving the screen, which is the content of Lemma 1. A fixed M must dominate it, and setting M at the mean residual violation (0.043) undercovers, while nominal coverage is reached near the ninety-fifth percentile (0.057) and well below the maximum (0.112). The recommendation follows, to calibrate M to the residual violation of a typical selected set, not its average, and treat it as a floor.

Finally, we repeat the estimand demonstration on generated panel data, where individual outcomes are simulated for a staggered design with a level confound and idiosyncratic noise, cohort event-study coefficients are estimated from genuine difference-in-differences contrasts against a never-treated group, and their covariance is estimated rather than supplied. The conclusions are unchanged, since across the informativeness axis the ATT estimator’s bias stays flat near $+0.30$ while the LATT’s falls toward zero, reproducing Figure 1 with estimated quantities, and the estimated standard error matches its Monte Carlo counterpart where selection is effectively deterministic. One discrepancy is informative. At intermediate informativeness, where selection is stochastic, the estimated standard error understates its Monte Carlo counterpart (0.016 against 0.024 at $\phi = 0.75$), the missing component being the variance induced by the randomness of selection, which a standard error conditional on the realized set does not capture. This confirms, from the panel side, that conditioning on a data-dependent selection understates uncertainty, and why the honest object to report is the sensitivity interval of Section 3.4 rather than a conventional confidence interval.

4 Empirical application

We illustrate the method with the local effect of the shale boom on house prices, a staggered design with many counties per cohort. The treatment date is the year a county’s oil and gas production first rises sharply, from county production records (U.S. Department of Agriculture, Economic Research Service, 2015), which sorts the boom counties into onset cohorts, and the outcome is the log county house-price index (Bogin et al., 2019), over 1998–2019. We estimate Callaway and Sant’Anna (2021) group-time effects against never-treated counties, those with negligible production, screen cohorts on the flatness of their estimated pre-trend, and report the level bound of Section 2 for the selected aggregate.

Onset is gradual and partly endogenous, which is what makes the setting instructive. A county’s production ramps up over several years, and where it does so during a regional housing upswing its prices are already climbing before the boom is dated. Figure 3(a) shows the result. The early cohorts have flat pre-trends and are retained, while the later cohorts, whose onset coincides with

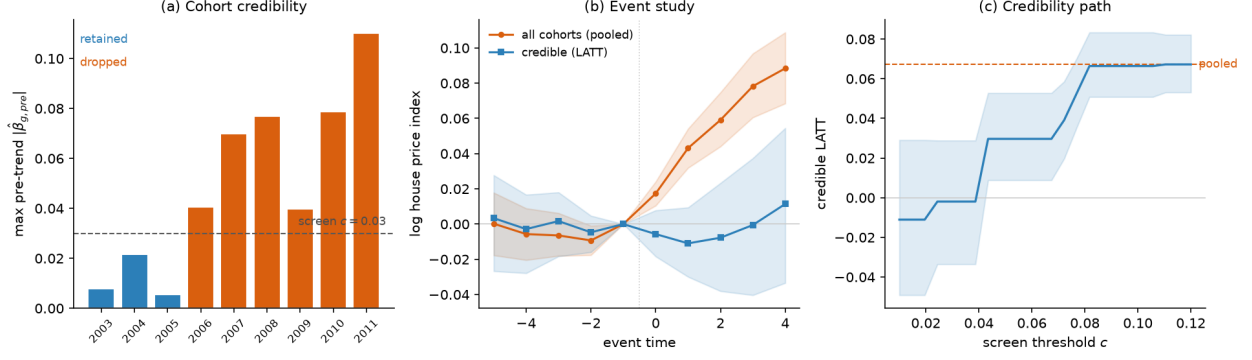


Figure 3: The shale boom and county house prices. (a) Cohort credibility map, the maximum pre-treatment violation by onset cohort with the flatness screen, where the later, trending cohorts are dropped. (b) Event study, all cohorts (pooled estimate) versus the credible subpopulation (LATT) at the illustrative threshold $c \approx 0.03$, with 95% bands. (c) The credibility path, the LATT as the screen threshold c is relaxed, near zero across the flat-pre-trend cohorts and rising toward the pooled estimate only as trending cohorts are readmitted.

the mid-2000s house-price run-up, carry pre-treatment trends several times larger and are dropped.

The two aggregates part ways (Figure 3b). Pooled across all cohorts, the event study climbs through the onset date to a house-price effect of +6.7 log points ($t = 8$), the kind of estimate a practitioner would report as a clear positive effect of the boom. But the pooled path is already trending before treatment, and the effect is inherited from the dropped cohorts. Restricted to the credible subpopulation, the three cohorts whose pre-trends are essentially flat (the screen at $c \approx 0.03$), the event study is flat both before and after onset, and the LATT is -0.2 log points, indistinguishable from zero. The gap between the two, +6.9 log points, is itself sharply estimated ($t = 5$). Honest inference confirms the reading rather than resting on the point estimate. Applying the level bound of Section 2 to the credible aggregate at M equal to the screen tolerance $c \approx 0.03$, the largest residual post-treatment violation the screen leaves room for, the LATT's fixed-length interval runs from -5.9 to $+5.5$ log points, which contains zero and excludes the pooled +6.7. The breakdown value is $M^* \approx 4.2$ log points, so reconciling the credible near-zero reading with the pooled effect would require countenancing a post-treatment violation about 1.4 times the screen tolerance and larger than any retained cohort's own pre-trend. The near-zero reading is not an artifact of a hand-picked threshold. Panel (c) traces the estimate along the credibility path of Section 2.3, and it stays flat and close to zero across the whole range of genuinely flat pre-trends, cohorts with $\max_e |\hat{\beta}_{g,pre}(e)|$ below about 0.03, and climbs toward the pooled +6.7 only once the threshold is loosened enough to readmit cohorts whose own pre-trends, around 0.07–0.08, are as large as the effect in question. Reading the LATT at that lax end simply rebuilds the pooled estimate out of its least credible cohorts, and the positive figure survives only by abandoning the credibility standard, not by tolerating a residual violation of the credible set.

The reading is that the pooled effect reflects a pre-existing trend in the counties that adopt late rather than a credibly causal effect, and a near-zero net effect for the credible subpopulation is

what the offsetting amenity and disamenity channels of local energy development would predict (Muehlenbachs et al., 2015). The method identifies the effect for the credible cohorts alone, and the excess in the dropped cohorts blends their steep pre-trends with any genuine difference in their treatment effects, a mixture the design cannot separate and need not, since the credible reading stands on the retained cohorts. The application thus shows the method working in the direction opposite to point-identification, not recovering an effect the pooled estimate misses but withdrawing one it spuriously reports, and locating that withdrawal exactly in the cohorts the screen flags.

5 Conclusion

We have argued that when parallel trends fails for some treated cohorts, a productive response is to change the estimand rather than to defend or bound the ATT. The credible-subpopulation LATT is point-identified under parallel trends for the selected cohorts alone, a condition that can hold when the ATT’s fails, is estimated by a transparent reweighting of standard group-time effects, and can be accompanied by honest sensitivity bounds on the residual violation that the pre-trend screen cannot rule out. This is a specific instance of the general principle, familiar from instrumental variables and limited overlap, of retreating to a credibly-identified subpopulation.

The method delivers a sharp answer where the ATT estimator is biased and its sensitivity set is wide, at the price of a narrower, subpopulation target whose composition must be described rather than assumed. Its advantage is contingent on pre-trends being informative about post-treatment violations. Where they are not, only economic context can carry the assumption. Under data-driven selection the estimator carries a pre-test bias that vanishes as pre-trends grow informative and a pointwise, rather than uniform, guarantee, which a researcher can trade for uniform validity by selecting on independent information. The principal open question is a decision-theoretic characterization of the region in which the point-identified LATT dominates the set-valued ATT, which would turn the scope condition documented here into a formal recommendation.

References

- Armstrong, Timothy B. and Michal Kolesár**, “Optimal Inference in a Class of Regression Models,” *Econometrica*, 2018, *86* (2), 655–683.
- Bilinski, Alyssa and Laura A. Hatfield**, “Nothing to See Here? Non-inferiority Approaches to Parallel Trends and Other Model Assumptions,” *arXiv preprint arXiv:1805.03273*, 2020.
- Bogin, Alexander N., William M. Doerner, and William D. Larson**, “Local House Price Dynamics: New Indices and Stylized Facts,” *Real Estate Economics*, 2019, *47* (2), 365–398.
- Borusyak, Kirill, Xavier Jaravel, and Jann Spiess**, “Revisiting Event-Study Designs: Robust and Efficient Estimation,” *Review of Economic Studies*, 2024. Forthcoming.
- Callaway, Brantly and Pedro H. C. Sant’Anna**, “Difference-in-Differences with Multiple Time Periods,” *Journal of Econometrics*, 2021, *225* (2), 200–230.
- Crump, Richard K., V. Joseph Hotz, Guido W. Imbens, and Oscar A. Mitnik**, “Dealing with Limited Overlap in Estimation of Average Treatment Effects,” *Biometrika*, 2009, *96* (1), 187–199.
- de Chaisemartin, Clément**, “Using Pre-Trends for Inference in Difference-in-Differences,” *arXiv preprint arXiv:2607.21312*, 2026.
- **and Xavier D’Haultfoeulle**, “Two-Way Fixed Effects Estimators with Heterogeneous Treatment Effects,” *American Economic Review*, 2020, *110* (9), 2964–2996.
- Fithian, William, Dennis Sun, and Jonathan Taylor**, “Optimal Inference after Model Selection,” *arXiv preprint arXiv:1410.2597*, 2014.
- Freyaldenhoven, Simon, Christian Hansen, and Jesse M. Shapiro**, “Pre-event Trends in the Panel Event-Study Design,” *American Economic Review*, 2019, *109* (9), 3307–3338.
- Goodman-Bacon, Andrew**, “Difference-in-Differences with Variation in Treatment Timing,” *Journal of Econometrics*, 2021, *225* (2), 254–277.
- Imbens, Guido W. and Joshua D. Angrist**, “Identification and Estimation of Local Average Treatment Effects,” *Econometrica*, 1994, *62* (2), 467–475.
- Kahn-Lang, Ariella and Kevin Lang**, “The Promise and Pitfalls of Differences-in-Differences: Reflections on *16 and Pregnant* and Other Applications,” *Journal of Business and Economic Statistics*, 2020, *38* (3), 613–620.
- Kwon, Soonwoo and Jonathan Roth**, “(Empirical) Bayes Approaches to Parallel Trends,” *AEA Papers and Proceedings*, 2024, *114*, 606–609.

- Leeb, Hannes and Benedikt M. Pötscher**, “Model Selection and Inference: Facts and Fiction,” *Econometric Theory*, 2005, *21* (1), 21–59.
- **and** –, “Sparse Estimators and the Oracle Property, or the Return of Hodges’ Estimator,” *Journal of Econometrics*, 2008, *142* (1), 201–211.
- Li, Fan, Kari Lock Morgan, and Alan M. Zaslavsky**, “Balancing Covariates via Propensity Score Weighting,” *Journal of the American Statistical Association*, 2018, *113* (521), 390–400.
- Manski, Charles F. and John V. Pepper**, “How Do Right-to-Carry Laws Affect Crime Rates? Coping with Ambiguity Using Bounded-Variation Assumptions,” *Review of Economics and Statistics*, 2018, *100* (2), 232–244.
- Muehlenbachs, Lucija, Elisheba Spiller, and Christopher Timmins**, “The Housing Market Impacts of Shale Gas Development,” *American Economic Review*, 2015, *105* (12), 3633–3659.
- Rambachan, Ashesh and Jonathan Roth**, “A More Credible Approach to Parallel Trends,” *Review of Economic Studies*, 2023, *90* (5), 2555–2591.
- Roth, Jonathan**, “Pretest with Caution: Event-Study Estimates after Testing for Parallel Trends,” *American Economic Review: Insights*, 2022, *4* (3), 305–322.
- , **Pedro H. C. Sant’Anna, Alyssa Bilinski, and John Poe**, “What’s Trending in Difference-in-Differences? A Synthesis of the Recent Econometrics Literature,” *Journal of Econometrics*, 2023, *235* (2), 2218–2244.
- Sun, Liyang and Sarah Abraham**, “Estimating Dynamic Treatment Effects in Event Studies with Heterogeneous Treatment Effects,” *Journal of Econometrics*, 2021, *225* (2), 175–199.
- U.S. Department of Agriculture, Economic Research Service**, “County-level Oil and Gas Production in the United States,” 2015. Onshore production, lower 48 states, 2000–2011.